

Prevalence and Outcomes of Anemia in Rheumatoid Arthritis: A Systematic Review of the Literature

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Anemia is a common comorbidity in individuals with rheumatoid arthritis (RA). In fact, anemia of the type characterized by low serum iron concentrations in conjunction with adequate iron stores is frequently associated with RA and has served as a model for anemia of chronic disease. A systematic search of the scientific literature published since January 1966 identified 19 articles that reported findings on either the prevalence of anemia in patients with RA or outcomes for patients with anemia and RA. Ten articles addressed the prevalence of anemia in patients with RA. Estimates of the prevalence of mild anemia ranged between 33% and 60%; however, the 2 studies that examined demographics in patients with RA did not identify subpopulations at particular risk for anemia. Twelve articles assessed the impact of the resolution of anemia on symptoms and quality of life (QOL) in patients with RA. For many of the parameters assessed—including swollen, painful, and tender joints, pain, muscle strength, and energy levels—a positive correlation was observed between improvement of symptoms and the resolution of anemia. In addition, 2 studies reported a significant improvement in QOL scores in patients with RA who experienced a response to treatment for anemia. These results suggest that (1) patients with RA who have anemia are likely to have more severe joint disease and (2) if the anemia is successfully treated, the joint disease will likely respond to treatment as well. Whether improvements in QOL and/or joint symptoms occur with improvement of anemia, independent of other signs of an overall response to RA therapy, remains to be determined. *Am J Med.* 2004; 116(7A):50S–57S. © 2004 by Excerpta Medica, Inc.

Rheumatoid arthritis (RA) is an autoimmune disease characterized by symmetric chronic inflammation of small and large joints. Destruction of cartilage and periarticular bone results from the release of neutral proteases and matrix metalloproteinases as part of the inflammatory process within the synovium. Although RA originates in the synovial tissues of the joints, it is a systemic disease that in its more severe form may include rapidly progressive multisystem inflammation.¹ The most common symptoms of RA manifested in the joints are inflammation and swelling, which are usually accompanied by pain, discomfort, and periods of prolonged morning stiffness. Additional common symptoms that are more systemic in nature include loss of muscle strength and low general energy levels. Any and all of these symptoms are likely to have a negative impact on the quality of life (QOL) of patients with RA.

RA has been estimated to occur in approximately 1% of the adult US population and is 3 times more prevalent in women than in men; there are approximately 30,000 to 50,000 prevalent (active and inactive) cases of juvenile RA.^{2,3} The etiology of the disease appears to be multifactorial—with genetic, environmental, and endocrinologic factors all playing roles.^{4,5}

There is no cure for RA. Current treatments focus on slowing disease progression and providing symptomatic relief.¹ Disease-modifying antirheumatic drugs (e.g., methotrexate) and biologic response modifiers, a class of drugs intended to inhibit inappropriate cytokine activity, have been used with substantial efficacy to alter the progression of RA by targeting cells and molecules that are thought to be pathophysiologically important. Nonsteroidal anti-inflammatory drugs (NSAIDs), such as cyclooxygenase-2 inhibitors, are often used to treat the symptomatic inflammation associated with RA.

Although some cases of anemia associated with RA can be attributed to iron deficiency caused by gastrointestinal bleeding associated with the use of NSAIDs, most cases of anemia ($\geq 60\%$) occurring in patients with RA are characterized as anemia of chronic disease and, in fact, have served as a model for the condition known as anemia of chronic disease.⁶ The increased production of inflammatory cytokines (e.g., tumor necrosis factor- α) observed in RA is linked to a cytokine-mediated decrease in erythro-

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This article was sponsored by the National Anemia Action Council, Inc., and funded by an educational grant from Amgen, Inc.

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